📘 3.4 Zero-Crossing Edge Detector

🧠 What is it?

A **zero-crossing edge detector** is a technique used in image processing to detect edges by identifying the points where the **second derivative** of the image intensity function changes sign (i.e., crosses zero). This change in sign is called a **zero-crossing**, and it usually corresponds to an edge in the image.

✍️ Key Concepts

➤ What is a Zero-Crossing?

- A **zero-crossing** occurs at a point where a function (e.g., second derivative of intensity) changes sign — from positive to negative or negative to positive.
- In edge detection, these points are likely to represent **edges or boundaries** in an image.

➤ Why Use the Second Derivative?

- The **first derivative** detects where the intensity changes the most (gradients).
- The **second derivative** tells us where those changes occur most rapidly — and zero-crossings of the second derivative often line up with actual **edges** in the image.

🧪 Example Workflow

1. **Start with the input image** (a 2D array of pixel intensities).
2. **Compute the first derivative** to measure gradient strength (not used directly here but good for comparison).
3. **Compute the second derivative** to find points where the gradient changes direction.
4. **Find zero-crossings** in the second derivative — these indicate potential edges.

🧮 Laplacian Operator

The **Laplacian** operator is a second-order differential operator used to compute the second derivative of an image.

Formula:

$$\nabla^2 I(x,y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

This means the Laplacian measures how much the image intensity at a pixel differs from its neighbors.

📏 Discrete Laplacian (for digital images)

In image processing, we approximate the Laplacian using **finite differences**:

- Second-order derivative in x-direction:

$$\frac{\partial^2 I(i,j)}{\partial x^2} \approx I(i,j+1) - 2I(i,j) + I(i,j-1)$$

- Second-order derivative in y-direction:

$$\frac{\partial^2 I(i,j)}{\partial y^2} \approx I(i+1,j) - 2I(i,j) + I(i-1,j)$$

Combined Laplacian Kernel (used in convolution):

$$\nabla^2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This kernel calculates the difference between a pixel and the average of its four immediate neighbors.

> ✅ **Remark: The zero-crossings of the Laplacian correspond to places where the image intensity curvature changes, which are often edges.**

🌊 Laplacian of Gaussian (LoG)

➤ What is LoG?

The **Laplacian of Gaussian (LoG)** combines **Gaussian smoothing** with the **Laplacian** operation to create a more robust edge detection method that handles noise better.

🧭 Steps in LoG:

1. **Apply Gaussian Smoothing**:
   - Reduces noise and small fluctuations in the image.
   - Controlled by the parameter **σ (sigma)**.
2. **Apply Laplacian Operator**:
   - Computes second derivatives on the smoothed image.
3. **Find Zero-Crossings**:

- Detect edges by identifying where the Laplacian output changes sign.

🧩 Discrete Zero-Crossing Detection

Because images are made of **discrete pixels**, we can't always detect a perfect zero. So we identify zero-crossings by checking **where the sign of neighboring pixel values changes**.

An edge pixel is:

- One where **neighboring pixel values** in the Laplacian result **change sign**.
- Preferably where the **absolute value of the magnitude** is smallest (closest to zero).